Rundong Luo

github.com/red-fairy | redfairy.github.io | rl897@cornell.edu

Education

Cornell University — Ithaca, NY, United States
Ph.D. in Computer Science — Aug. 2024 – Present
Advisors: Prof. Wei-Chiu Ma and Prof. Noah Snavely

Peking University — Beijing, China
Bachelor of Science in Computer Science and Technology (summa cum laude) — Aug. 2020 – Jun. 2024
Advisors: Prof. Yisen Wang and Prof. Jiaying Liu

Research Experience

Cornell Vision and Graphics Lab — Aug. 2024 – Present
Advisors: Prof. Wei-Chiu Ma and Prof. Noah Snavely — Cornell University

- Research on 3D computer vision, focusing on 3D/video understanding and generation; contributed to new methods for panoramic video synthesis and computational art.

Stanford Vision and Learning Lab — Jan. 2023 – Oct. 2023
Advisor: Prof. Jiajun Wu — Stanford University

- Research on 3D computer vision, specifically object-centric learning in 3D scenes with neural fields.
- Published a co-first author paper at TMLR.

Self-Directed Research — Jan. 2024 – Jun. 2024
Peking University

- Research on articulated object generation.
- Published a co-first author paper at ICRA 2025.

STRUCT Lab — Apr. 2022 – Jun. 2024
Advisor: Prof. Jiaying Liu — Peking University

- Research on improving vision systems performance in low-illumination environments;
- Published a first-author paper at ICCV 2023 (oral presentation, top 2%) and a co-first-author paper at TPAMI.

ZERO Lab — Jul. 2021 – Sept. 2022
Advisor: Prof. Yisen Wang — Peking University

- Research on adversarial machine learning and self-supervised learning; investigated self-supervised defense strategies against model vulnerabilities.
- Published a co-first author at ICLR 2023.

Publications and Manuscripts

* indicates equal contributions

- **Rundong Luo**, Noah Snavely, and Wei-Chiu Ma. ShadowDraw: From Any Object to Shadow–Drawing Compositional Art. Under review, 2025.
- **Rundong Luo**, Matthew Wallingford, Ali Farhadi, Noah Snavely, and Wei-Chiu Ma. Beyond the Frame: Generating 360° Panoramic Videos from Perspective Videos. In ICCV, 2025.
- **Rundong Luo***, Haoran Geng*, Congyue Deng, Puhao Li, Zan Wang, Baoxiong Jia, Leonidas Guibas, and Siyuan Huang. PhysPart: Physically Plausible Part Completion for Interactable Objects. In ICRA, 2025.
- **Rundong Luo***, Hong-Xing Yu*, and Jiajun Wu. Unsupervised Discovery of Object-Centric Neural Fields. TMLR, 2025.
- Wenjing Wang*, **Rundong Luo***, Wenhan Yang, and Jiaying Liu. Unsupervised Illumination Adaptation for Low-Light Vision. TPAMI, 2024.
- **Rundong Luo**, Wenjing Wang, Wenhan Yang, and Jiaying Liu. Similarity Min-Max: Zero-shot Day-Night Domain Adaptation. In ICCV, 2023.
- **Rundong Luo***, Yifei Wang*, and Yisen Wang. Rethinking the Effect of Data Augmentation in Adversarial Contrastive Learning. In ICLR, 2023.

Academic Service

- Workshop Organizer: Synthetic Data in Computer Vision, CVPR 2026.
- Reviewer: CVPR 2024–2026, NeurIPS 2024-2025, WACV 2025, ICLR 2025-2026, ICCV 2025, ICML 2025, 3DV 2026, TCSVT.
- Teaching Assistant: 3D Vision (Cornell, 2024 Fall), Practice of Programming in C&C++ (PKU, 2023 Spring).

Selected Honors and Awards

- Outstanding Undergraduate Thesis (Beijing Municipal, 190 awarded at PKU), 2024.
- Excellent Undergraduate Graduate (Beijing Municipal, 34 awarded at PKU), 2024.
- Sensetime Scholarship (awared to 30 AI undergraduates nationwide), 2024.
- Chinese National Scholarship (awarded to top 0.2% undergraduates in China), 2023.